

Output tables for 1xN statistical comparisons.

March 8, 2025

1 Average rankings of Friedman test

Average ranks obtained by each method in the Friedman test.

Algorithm	Ranking
Random	8.875
NN	7.5
Sweep	9
SaveSeq	2.875
SaveParall	1.875
MJ	2.375
CMT	2.875
Kilby	6
SaveMatch	5.625
FJ	8

Table 1: Average Rankings of the algorithms (Friedman)

Friedman statistic (distributed according to chi-square with 9 degrees of freedom): 61.827273.
P-value computed by Friedman Test: 0.

2 Post hoc comparison (Friedman)

P-values obtained in by applying post hoc methods over the results of Friedman procedure.

i	algorithm	$z = (R_0 - R_i)/SE$	p	Holm	Finner	Li
9	Sweep	4.70662	0.00003	0.005556	0.005683	0.013622
8	Random	4.624048	0.00004	0.00625	0.011334	0.013622
7	FJ	4.046042	0.00052	0.007143	0.016952	0.013622
6	NN	3.715753	0.000203	0.008333	0.022539	0.013622
5	Kilby	2.724885	0.006432	0.01	0.028094	0.013622
4	SaveMatch	2.477168	0.013243	0.0125	0.033617	0.013622
3	SaveSeq	0.660578	0.508883	0.016667	0.039109	0.013622
2	CMT	0.660578	0.508883	0.025	0.04457	0.013622
1	MJ	0.330289	0.741182	0.05	0.05	0.05

Table 2: Post Hoc comparison Table for $\alpha = 0.05$ (FRIEDMAN)

Holm's procedure rejects those hypotheses that have an unadjusted p-value ≤ 0.0125 .

Finner's procedure rejects those hypotheses that have an unadjusted p-value ≤ 0.039109 .

Li's procedure rejects those hypotheses that have an unadjusted p-value ≤ 0.013622 .

3 Adjusted P-Values (Friedman)

Adjusted P-values obtained through the application of the post hoc methods (Friedman).

i	algorithm	unadjusted p	p_{Holm}
1	Sweep	0.00003	0.000023
2	Random	0.00004	0.00003
3	FJ	0.00052	0.000365
4	NN	0.000203	0.001216
5	Kilby	0.006432	0.032162
6	SaveMatch	0.013243	0.052972
7	SaveSeq	0.508883	1.526648
8	CMT	0.508883	1.526648
9	MJ	0.741182	1.526648

Table 3: Adjusted p -values (FRIEDMAN) (I)

i	algorithm	unadjusted p	p_{Finner}	p_{Li}
1	Sweep	0.00003	0.000023	0.00001
2	Random	0.00004	0.000023	0.000015
3	FJ	0.00052	0.000156	0.000201
4	NN	0.000203	0.000456	0.000782
5	Kilby	0.006432	0.011548	0.02425
6	SaveMatch	0.013243	0.019798	0.048676
7	SaveSeq	0.508883	0.599177	0.662866
8	CMT	0.508883	0.599177	0.662866
9	MJ	0.741182	0.741182	0.741182

Table 4: Adjusted p -values (FRIEDMAN) (II)